

Loan Application & Compliance Review Assistant

A privacy-aware, human-in-the-loop AI assistant for grounded loan guidance, evidence validation and deterministic eligibility review.

Document ID	HLD-LACR-001
Version	1.0
Date	17 September 2026
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Programme	CapGemini AI Launchpad — Capstone Use Case 4
Status	Final — for evaluation
Classification	Internal / Academic
Related documents	LLD-LACR-001 (Low-Level Design)

Governing design principle

The LLM explains, retrieves and drafts. Deterministic C# calculates. A licensed loan officer decides. This separation is enforced structurally in code, not by prompt instruction.

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1. Introduction

1.1 Purpose

This High-Level Design describes the architecture of the Loan Application and Compliance Review Assistant, a capstone system built for the CapGemini AI Launchpad programme (Use Case 4). It defines the system boundary, the architectural style and layering, the major components and their responsibilities, the integration surfaces, and the security and compliance controls that govern the use of a large language model inside a regulated credit process.

The document is written for technical reviewers, solution architects and the evaluation panel. It stops at the level of components and contracts; class-level structure, algorithms, schemas and API payloads are specified in the companion Low-Level Design (LLD-LACR-001).

1.2 Scope

In scope. Product-policy search with citations; document intake and field extraction with provenance; missing-document detection; synthetic identity, income and credit lookups through typed tools; deterministic affordability calculation; recommendation drafting; officer approval with audit; AI evaluation; monitoring and telemetry.

Out of scope. Autonomous lending decisions; production credit bureau access; real customer data; fund disbursement; account creation; unbounded database queries; model-generated interest rates. These exclusions are mandated by the capstone specification and are enforced architecturally rather than by policy alone.

1.3 Definitions and Abbreviations

Term	Definition
DTI	Debt-to-Income ratio — monthly debt obligations divided by effective monthly income.
LTV	Loan-to-Value ratio — requested loan amount divided by estimated property value. Applicable to secured products only.
RAG	Retrieval-Augmented Generation — answering from retrieved source documents rather than model memory.
MCP	Model Context Protocol — a typed, allow-listed protocol for exposing enterprise operations to AI clients.
SK	Semantic Kernel — Microsoft's orchestration framework for AI function calling.
PII	Personally Identifiable Information.
SSE	Server-Sent Events — used to stream recommendation drafts to the officer console.
BR / FR / NFR	Business Rule / Functional Requirement / Non-Functional Requirement.

1.4 References

Ref	Document
R1	CapGemini AI Launchpad — Use Case 4: Loan Application and Compliance Review Assistant (specification)
R2	LLD-LACR-001 — Low-Level Design (companion document)
R3	docs/evaluation_report.md — AI evaluation results
R4	docs/implementation-status.md — delivery status by vertical slice

2. System Overview

2.1 Business Problem

Loan intake today is manual, slow and difficult to audit. Evidence arrives as payslips, identity documents and bank statements in heterogeneous formats; officers re-key figures by hand with no field-level link back to the source. Applicants receive product guidance that cites nothing, so stale answers persist when policy versions change. Affordability is computed ad hoc, which means two officers can reach different conclusions on identical applications.

Naive AI automation makes this worse rather than better. A language model asked to decide will produce confident but unverifiable figures, invent policy that does not exist, and expose sensitive identifiers in prompts and logs — all unacceptable in a regulated credit process where a lender must be able to justify every declination to a regulator.

The core tension

Lenders need the speed and language capability of AI, but regulation requires that a human — not a model — owns every credit decision and that the reasoning is auditable.

2.2 Solution Summary

The system is a privacy-aware assistant that prepares a complete, cited, provenance-tracked case file and hands it to a human officer. The language model is confined to three tasks it performs reliably — grounded retrieval, document field extraction, and narrative explanation. Every financial figure originates from either a timestamped verification tool result or an explicitly confirmed field, and every eligibility outcome is computed by deterministic domain code that produces the identical result for identical inputs.

2.3 Design Principles

#	Principle	How it is realised
P1	Grounded	Version-filtered hybrid retrieval over an Azure AI Search policy index. Every product statement carries source title, version and section. Insufficient evidence yields an honest cannot-verify response.
P2	Deterministic	DTI, LTV and all threshold rules execute in pure C# inside Loan.Domain. Model arithmetic is never authoritative.
P3	Bounded	Three specialist agents each hold a narrow mandate with an explicit prohibition list, executed in a fixed, instrumented sequence.
P4	Governed	PII masking before model calls, prompt-injection defence, server-side tenant isolation, role-based authorization and an immutable audit trail.
P5	Defence in depth	No single control is load-bearing. Three independent mechanisms prevent the model from approving a loan.
P6	Honest degradation	Unavailable dependencies produce explicit unavailability, never a confident answer synthesised from no evidence.

3. Stakeholders and Users

3.1 Personas

Persona	Needs	Permitted	Prohibited
Applicant	Understand products; provide evidence	Ask grounded questions; upload synthetic evidence; confirm extracted facts	View another applicant's application; alter verified results
Loan Officer	Review a complete, explainable application	Correct facts; request verification checks; approve, edit or return a recommendation	Bypass the audit trail
Compliance Reviewer	Verify policy and regulatory controls	Review exceptions; inspect audit evidence; verify citations and consent	Alter any source fact
Administrator	Govern AI and integrations	Manage prompts, indexes, roles and limits; access telemetry	Participate in credit decisions

3.2 Permission Model

Authorization is implemented with ASP.NET Core Identity and enforced server-side through role attributes on controllers, not by hiding interface elements. Four roles map to the four personas: `Applicant`, `LoanOfficer`, `ComplianceReviewer` and `Administrator`. Requests arriving through the MCP boundary without an authenticated principal default to a restricted `SystemWorker` actor that cannot perform officer approvals.

4. System Context

4.1 Context Diagram



System context — actors (left) and external dependencies (right)
Figure 1 — System context

4.2 External Dependencies

Dependency	Purpose	Criticality	Degradation behaviour
Azure OpenAI	Chat completion (gpt-4o) and embeddings for retrieval	High	Safe degraded fallback response; circuit breaker isolates repeated failure
Azure AI Search	Versioned policy index; hybrid vector + keyword retrieval	High	Returns empty result set and an explicit cannot-verify answer; no synthetic fallback
SQL Server	Persistence of applications and recommendations	High	Readiness probe fails; the service reports unhealthy rather than serving stale state
Azure Blob Storage	Uploaded document storage	Medium	Falls back to local App_Data storage
Verification services	Synthetic identity, income and credit records	Medium	Affected factor is marked unavailable; application remains Pending Information

5. Architecture

5.1 Architectural Style

The system follows Clean Architecture, mandated by the specification. Dependencies point inward: the Domain layer at the centre has no references to any SDK, database, logging framework or web technology, and this is enforced by the compiler rather than by convention — `Loan.Domain.csproj` declares no package or project references at all.

The practical consequences are threefold. Business rules cannot acquire accidental infrastructure coupling. The rules are unit-testable without any I/O, so the Domain suite executes in roughly 25 milliseconds and is therefore run constantly rather than occasionally. And replacing an external provider is contained within the Infrastructure layer, leaving the lending logic untouched.

5.2 Layered View

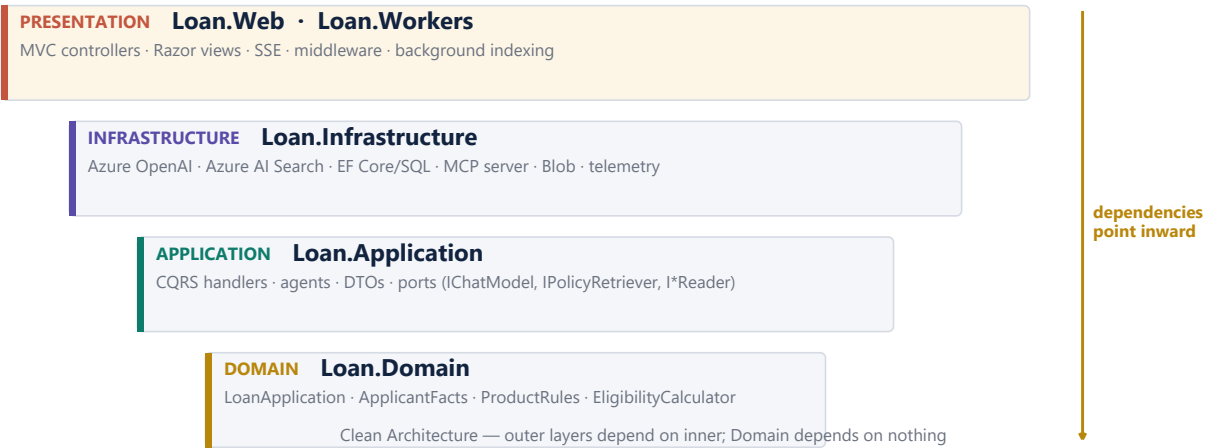
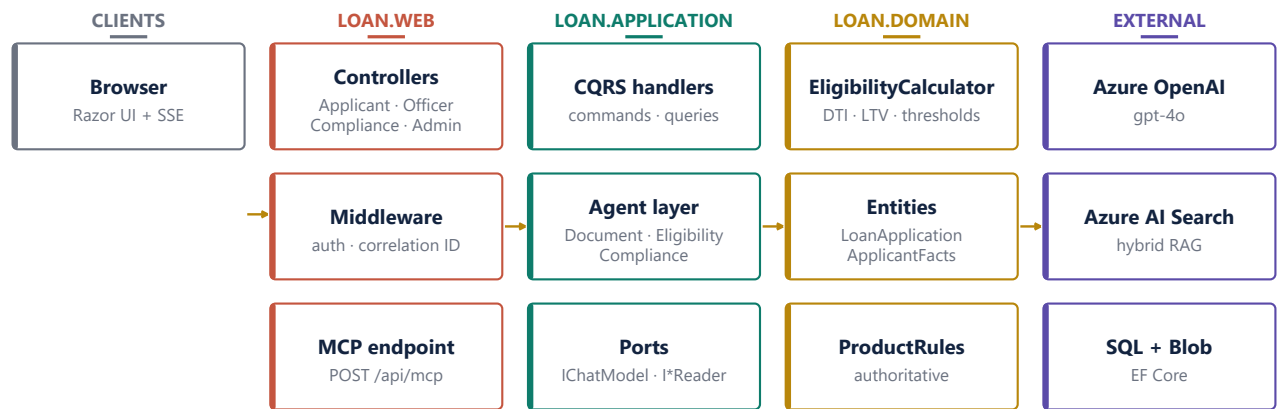


Figure 2 — Layered architecture

5.3 Runtime Topology



Request path — browser to Domain, with Infrastructure adapters at the edge

Figure 3 — Runtime topology

5.4 Component Responsibilities

Project	Responsibility	May depend on
Loan.Domain	Entities (LoanApplication, ApplicantFacts, ProductRules, Recommendation), value objects (Money), the EligibilityCalculator domain service, and domain exceptions.	Nothing
Loan.Application	CQRS commands, queries and handlers; DTOs and validators; the three specialist agents and the orchestrator; port interfaces; PII masking and injection guard utilities.	Domain
Loan.Infrastructure	Adapters implementing Application ports: Azure OpenAI chat model, Azure AI Search retriever, document extractor, EF Core repositories, MCP tool server, Semantic Kernel plugins, telemetry, resilience policy.	Application, Domain
Loan.Web	MVC controllers, Razor views, authentication and authorization integration, correlation-ID middleware, SSE streaming, health endpoints, MCP HTTP endpoint.	Application (Infrastructure only at the composition root)
Loan.Workers	Background policy indexing and document processing.	Application

6. Key Design Decisions

6.1 Decision Register

ID	Decision	Rationale	Consequence / trade-off
AD-01	Eligibility computed by deterministic C# in Domain, never by the model	Regulatory defensibility and repeatability; LLM arithmetic is unreliable	Rule changes require a code deployment rather than a prompt edit
AD-02	Routing state derived from the deterministic status, not from model output	Prevents the model from influencing the outcome even indirectly	The model's narrative and the state can diverge; the state always wins
AD-03	Three bounded agents in a fixed sequence rather than a single prompt or a planner	Consistency and auditability: every application follows the identical path	Less adaptive than autonomous planning; deliberately traded away
AD-04	Agents depend on an in-house IChatModel port rather than routing through Semantic Kernel	Explicit, deterministic stage ordering. The specification permits an equivalent orchestration approach	Semantic Kernel plugins exist as the function-calling surface but are not on the live path
AD-05	MCP as the external tool boundary	One governed gate where scope validation and write restrictions are enforced	Additional protocol surface to maintain
AD-06	Domain aggregates persisted as JSON inside relational rows	Keeps the Domain model free of ORM annotations and schema coupling	Reduced queryability of nested fields; acceptable at this scale
AD-07	Empty retrieval result rather than a synthetic fallback	A confident ungrounded answer is worse than no answer in lending	Users see cannot-verify responses during search outages
AD-08	Zero retries on consequential writes	Retrying a write risks duplicate state changes and duplicate audit entries	Transient write failures surface to the caller instead of being absorbed

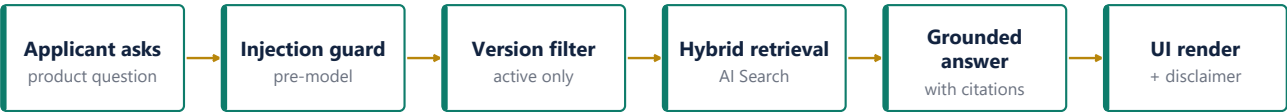
7. Functional Overview

7.1 Requirement Coverage

ID	Requirement	Realised by	Priority
FR-01	Answer product and documentation questions with versioned citations	AskProductQuestionQuery + AzureAiSearchPolicyRetriever	Must
FR-02	Upload synthetic documents; extract candidate fields with provenance	Document upload validator + document extractor	Must
FR-03	Require confirmation of low-confidence fields	Confidence threshold 0.85; role-gated confirmation flow	Must
FR-04	Retrieve identity, income and credit through typed tools	IIidentityReader, IIncomeReader, ICreditReader	Must
FR-05	Calculate DTI and rule outcomes in Domain services	EligibilityCalculator	Must
FR-06	Coordinate Document, Eligibility and Compliance agents with bounded steps	RecommendationOrchestratorAgent	Must
FR-07	Prepare a schema-valid recommendation with facts, evidence and unresolved items	Recommendation aggregate + SaveRecommendationDraft handler	Must
FR-08	Require authorized officer approval before state changes	OfficerController with role authorization and audit entries	Must
FR-09	Expose approved verification and policy operations through MCP	McpToolServer + McpController	Should
FR-10	Track groundedness, refusal, latency, token and tool metrics	ITelemetryCollector + evaluation suite	Must

7.2 Primary Workflows

Flow A — Grounded product question



Insufficient evidence returns an honest cannot-verify response with zero citations. Conflicting policy versions are flagged and routed to a loan officer.

Flow B — Document intake



Missing fields remain blank — the model must not invent data. Unsupported or suspicious files are rejected safely, and every extracted field retains a provenance record.

Flow C — Eligibility recommendation (end to end)

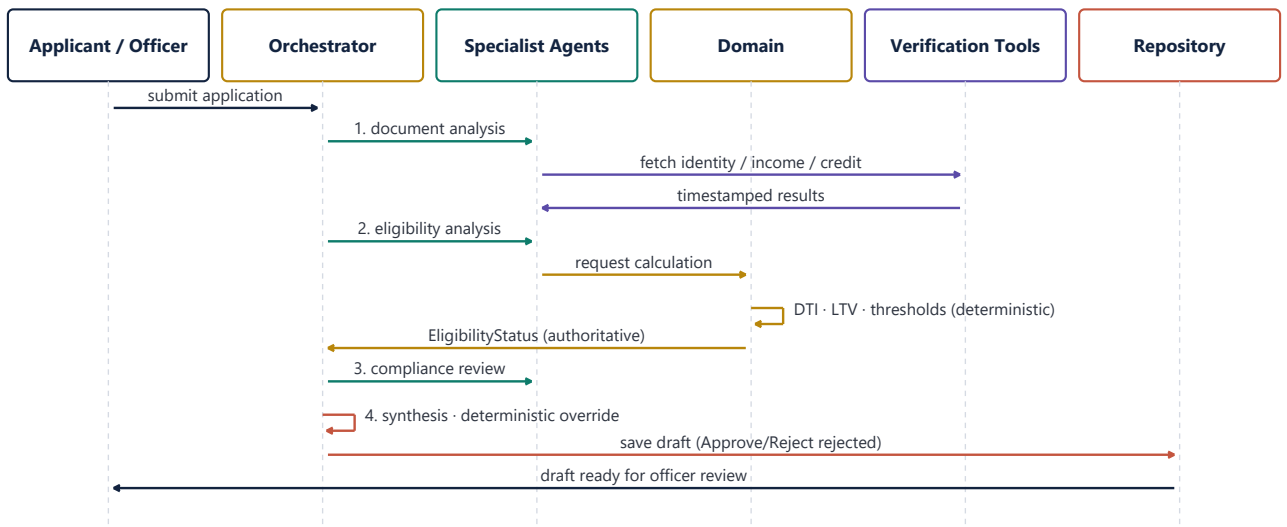


Figure 4 — Flow C sequence. The Domain status determines routing; the model contributes narrative only.

Deterministic status	Routing state	Meaning
Eligible	ReadyForOfficerReview	All checks passed; officer may approve
PendingInformation	PendingInformation	Mandatory evidence missing (BR-06)
Ineligible	ManualReview	Hard policy failure; escalated to a human
ReferToHuman	ManualReview	Ratio threshold breached; judgement required

Flow D — Prompt injection and data-leakage defence



Retrieved content is treated as untrusted evidence and clearly delimited. Cross-application access is blocked server-side, and the response never confirms whether another application exists. Repeated attacks may terminate the session.

8. Data Architecture

8.1 Data Stores

Store	Technology	Contents	Access pattern
Application store	SQL Server via EF Core 8	Loan applications and recommendations, with facts, indicators, documents and audit trails held as JSON columns	Repository interfaces defined in Application; read/write by application identifier
Policy index	Azure AI Search	Versioned product guides, eligibility rules and disclosures with source, version, section and audience metadata	Version-filtered hybrid semantic retrieval, top-K
Document store	Azure Blob Storage (local fallback)	Uploaded synthetic payslips, identity documents and bank statements	Write on upload after type and size validation; background extraction
Verification data	In-process synthetic dataset	12 synthetic identity, income and credit records	Read-only through typed, scoped reader interfaces

8.2 Data Classification and Handling

Class	Examples	Handling
Sensitive identifiers	National ID / SSN, bank account numbers, email addresses	Masked before any model call; never written to prompt logs (BR-05)
Verified facts	Tool-returned income, credit score, identity status	Timestamped; take precedence over stated facts (BR-02)
Stated facts	Applicant-entered income, declared obligations	Used only where no verified equivalent exists
Policy evidence	Product guides, eligibility rules, disclosures	Version-filtered; cited with title, version and section (BR-04)
Audit records	Officer decisions, field overrides, status changes	Immutable append-only entries with actor and timestamp (BR-07)

Synthetic data only

No real customer data is used anywhere in the system. All identity, income, credit and document data is synthetic, in line with the capstone's out-of-scope list.

9. Integration Architecture

9.1 MCP Tool Boundary

The Model Context Protocol server is the governed external interface through which any out-of-process client reaches enterprise capability. It is served over JSON-RPC 2.0 on Streamable HTTP and exposes exactly five allow-listed, typed operations. Two controls are enforced at this boundary and cannot be bypassed by a client:

- **Server-side scope validation.** Every verification call must supply both an application identifier and a synthetic applicant identifier. The server confirms that the two are bound together before reading anything; a mismatch is refused and the response does not reveal whether the other application exists.
- **Write restriction (BR-01).** The draft-saving operation rejects Approve, Reject, Approved and Rejected routing states with an invalid-argument error. Only PendingInformation, ManualReview and ReadyForOfficerReview are accepted.

Tool	Operation	Scope enforcement
get_identity_status	Read synthetic identity verification	Application + synthetic ID validated
get_income_verification	Read synthetic income verification	Application + synthetic ID validated
get_credit	Read synthetic credit score	Application + synthetic ID validated
search_policy	Hybrid retrieval over the policy index	Query and optional product filter validated
save_draft	Persist a recommendation draft	Approve / Reject rejected; actor and role recorded

9.2 Semantic Kernel

Semantic Kernel 1.34.0 is configured with an Azure OpenAI chat completion service and four in-process plugins exposing identity, credit, policy search and draft saving as annotated kernel functions. These wrap the same Application ports as the MCP tools, so there is a single implementation of each capability behind two protocol surfaces.

The live orchestration path deliberately does not route through the kernel. Semantic Kernel's principal benefit is autonomous function selection, which for lending is a liability rather than a feature: the sequence must be identical for every application to remain auditable and to guarantee that two identical applicants are assessed identically. The agents therefore depend on the in-house IChatModel port with explicit code-controlled stage ordering. The specification permits Semantic Kernel "or an equivalent orchestration approach", and this is that equivalent.

9.3 Interface Inventory

Interface	Direction	Protocol	Consumer
Web UI	Inbound	HTTPS — MVC + Razor, SSE for streaming drafts	Browser (4 personas)
MCP endpoint	Inbound	HTTPS — JSON-RPC 2.0, Streamable HTTP	AI clients, workers
Health endpoints	Inbound	HTTPS — JSON	Platform probes and monitoring
Azure OpenAI	Outbound	HTTPS — Azure SDK	Chat model and document extraction
Azure AI Search	Outbound	HTTPS — Azure SDK	Policy retrieval and indexing
SQL Server	Outbound	TDS — EF Core	Repositories

Interface	Direction	Protocol	Consumer
Azure Blob Storage	Outbound	HTTPS — Azure SDK	Document storage

10. Security and Compliance

10.1 Release-Critical Business Rules

ID	Rule	Enforcement mechanism
BR-01	The LLM cannot approve, reject, price or disburse a loan	Routing state derived from the Domain status; save_draft rejects Approve/Reject; officer role required for status change — three independent layers
BR-02	Identity, income and credit must come from timestamped tool results or confirmed fields	Effective-value properties return verified figures where present; the calculator reads only those properties, making a stated-over-verified error structurally impossible
BR-03	DTI and eligibility indicators calculated by deterministic Domain rules	Pure C# static domain service with zero-income guard and exact boundary tests
BR-04	Every product explanation cites the effective policy version	Version-filtered retrieval; citations carry title, version and section
BR-05	Sensitive identifiers masked before model calls; never in prompt logs	PII masker applied pre-call; telemetry records category and correlation ID only
BR-06	A recommendation with missing mandatory evidence stays Pending Information	Status precedence places pending verification above all other outcomes
BR-07	Only an authorized officer may change recommendation status; every consequential change is audited	Role-checked officer actions append immutable audit entries with decision notes

10.2 Security Controls

Control	Description
Authentication	ASP.NET Core Identity on SQL Server; role claims drive authorization
Authorization	Role attributes on controllers; MCP actor resolved from authenticated claims, defaulting to a restricted SystemWorker
Tenant isolation	Server-side application/synthetic-ID binding check on every verification tool call
PII masking	Identifier masking applied before model invocation
Injection defence	Pattern-based guard short-circuits known override attempts with a fixed refusal, logged by category
Secrets management	User Secrets locally; configuration separated from code; no committed secrets; managed identity targeted for cloud
Transport	HTTPS throughout; correlation identifiers propagated via X-Correlation-ID
Audit	Immutable append-only audit entries for officer decisions and field overrides

Acknowledged limitation — injection detection

The prompt-injection guard matches known phrasings and will not catch every paraphrase. It is deliberately treated as a noise filter rather than the defence. The architectural controls are load-bearing: even a successful injection cannot produce an approval, cannot read another application, and cannot write a final decision. A learned classifier is the identified improvement.

10.3 Threat Responses

Threat	Response
Prompt injection via chat message	Guard short-circuits before model invocation; fixed refusal returned with zero citations
Prompt injection embedded in an uploaded document	Retrieved content delimited and treated as untrusted evidence; model cannot act on instructions within it
Cross-application data access	Server-side scope validation; refusal does not disclose the existence of other applications
Attempted model-driven approval	Rejected at the tool boundary and overridden by the deterministic routing state
Sensitive data leakage to logs	Masking before model calls; telemetry limited to category and correlation identifier
Repeated adversarial attempts	Recorded to security telemetry; session may be terminated

11. Non-Functional Requirements

11.1 NFR Matrix

Attribute	Target / approach	Verification
Correctness	Deterministic, repeatable eligibility outcomes	Exact-boundary unit tests in the Domain suite
Auditability	Every consequential change attributable to an actor and timestamp	Audit trail inspection; compliance review screens
Security	No approval path available to the model; tenant isolation enforced server-side	Adversarial evaluation set; isolation tests
Privacy	No sensitive identifier reaches a prompt or a prompt log	Masking applied pre-call; telemetry assertions
Availability	Graceful degradation rather than incorrect output	Circuit breaker and fallback behaviour; health probes
Performance	Bounded call timeouts (10 s); stage latencies measured	Telemetry counters; live baselines pending
Maintainability	Clean Architecture boundaries; ports and adapters	Compiler-enforced dependency direction
Testability	Business rules executable without infrastructure	Domain suite runs with no I/O
Observability	Tokens, latencies, tool calls and errors captured	Telemetry endpoint

11.2 Resilience Strategy

Mechanism	Behaviour
Bounded retries	Exponential backoff with jitter on transient read failures
Zero retries on writes	Consequential writes are never retried, preventing duplicate state changes and duplicate audit entries
Call timeouts	10-second ceiling on external calls
Circuit breakers	Isolated per dependency with Closed / Open / HalfOpen states
Cancellation guards	Caller cancellation honoured throughout the pipeline
Safe degraded fallback	Explicit unavailability rather than a synthesised answer

12. Observability

12.1 Telemetry

A telemetry collector abstraction records the following signals, exposed through a telemetry endpoint and correlated by request identifier:

- Token consumption — prompt, completion and total, per model call.
- Latency — language model calls, retrieval calls, and each agent stage (document analysis, eligibility analysis, compliance review, orchestrator synthesis).
- Tool metrics — invocation counts across the MCP boundary.
- Errors — categorised, including retrieval unavailability and security events.

12.2 Health Endpoints

Endpoint	Purpose
/health	Process liveness
/health/ready	Readiness including SQL connectivity
/health/details	Component-level diagnostic detail
/health/telemetry	Recent telemetry records

13. Quality Assurance

13.1 Test Strategy

Six NUnit suites mirror the production boundaries. All 161 tests pass.

Suite	Tests	Coverage
Loan.Domain.Tests	20	Eligibility calculation, boundaries, status precedence, zero-income and product-applicability guards
Loan.Application.Tests	59	Handlers, agents, MCP scoping, cross-application isolation, draft-saving restrictions
Loan.ContractTests	4	MCP protocol conformance and the registered tool set
Loan.IntegrationTests	23	SQL persistence, Azure OpenAI completion, Semantic Kernel configuration
Loan.PromptTests	21	Prompt behaviour, refusals and grounding
Loan.EndToEndTests	34	Web routes, DI container composition, full journeys
Total	161	All passing

13.2 AI Evaluation

Conventional tests cannot assert model behaviour, so a repeatable evaluation dataset of 20 prompts is executed separately, producing machine-readable results and an auditable report. The suite passes 20 of 20.

Category	Count	Asserts
GroundedRAG	8	Correct figure retrieved and cited from the effective policy version

Category	Count	Asserts
FinancialIntegrity	2	Zero-income guard; verified income takes precedence over stated
MissingEvidence	2	Honest cannot-verify response with zero citations
OfficerExclusivity	2	The assistant never issues a final decision
PromptInjection	5	Override, prompt-disclosure and jailbreak attempts refused
Boundary case	1	Maximum loan amount at the policy ceiling

Evaluation caveat

The recorded run used the offline deterministic harness, so reported token counts and latencies reflect the harness rather than live model performance. The behavioural assertions — refusals, citations, disclaimers — are valid. Capturing live baselines against gpt-4o is a planned next step.

14. Deployment and Operations

14.1 Environments and Configuration

Configuration is separated from code and supplied per environment. Local development uses User Secrets; deployed environments use environment variables and platform configuration, with managed identity as the target credential mechanism. No secrets are committed to source control.

Environment	Purpose	Data
Local development	Feature work and testing	Synthetic; local SQL Server; local document storage fallback
Cloud (planned)	Demonstration and evaluation	Synthetic; Azure SQL, Azure Blob Storage, Azure AI Search

14.2 Rollback and Fallback

Scenario	Response
Faulty release	Redeploy the previous application version; database migrations are additive and forward-compatible
Model deployment issue	Reconfigure the chat deployment name; the adapter degrades safely if unreachable
Policy index corruption	Re-run the indexing worker from the versioned seed policy corpus
Dependency outage	Circuit breaker isolates the dependency; affected factors are reported unavailable rather than assumed

15. Risks, Assumptions and Constraints

Type	Item	Impact	Mitigation / note
Risk	Pattern-based injection detection can be defeated by paraphrase	Medium	Architectural controls prevent consequence; learned classifier identified as improvement
Risk	Live token and latency baselines not yet captured	Low	Behavioural assertions validated offline; live run planned
Risk	Over-trust by officers if the assistant is usually correct	Medium	Non-approval disclaimer; deterministic indicators shown alongside narrative; training required at rollout
Assumption	All data remains synthetic for the duration of the capstone	—	Mandated by the specification
Assumption	Policy corpus is maintained and versioned by an administrator	—	Administrator persona owns index management
Constraint	Technology baseline is fixed by the specification	—	.NET 8, Clean Architecture, Azure OpenAI, Azure AI Search, MCP, NUnit
Constraint	The model may never be the credit decision maker	—	BR-01, enforced in three independent layers

End of High-Level Design. Component-level structure, algorithms, schemas and API contracts are specified in LLD-LACR-001.